

Dated: _____

QUESTION NO. 2

```
DECLARE Type, Mem, Dis, Avail : CHARACTER
DECLARE SOC, Req SOC, Hour, Time, Charge Needed,
Rate, Charge Discount, Charge Cost, Base Parking,
Parking Discount, Parking Cost, Total Discount, Total
Payable : Real
DECLARE Priority, Time status, stay message : STRING
```

```
Print ("Enter Vehicle Type (E for Electric, H for Hybrid)")
```

```
INPUT Type
```

```
Print ("Enter Current Battery SOC (1-100):")
```

```
INPUT SOC
```

```
Print ("Enter Expected parking duration (Hours)")
```

```
INPUT Hours
```

```
Print ("Enter Current time (24 hours)")
```

```
INPUT Time
```

```
Print ("Do you have a Parking membership? (Y/N)")
```

```
INPUT mem
```

```
Print ("Do you have disabled member priority? (Y/N)")
```

```
INPUT Dis
```

```
Print ("Is charging station available? (Y/N)")
```

```
INPUT Avail
```

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IF Avail == "N"

THEN

IF Type == "H" THEN

Print ("Charging Unavailable, Parking only")

ELSE

Print ("No charging available")

ENDIF

Charge cost = 0.0

ELSE

IF NOT (Type == "E" OR (Type == "H" AND
SOC < 40))

THEN

Print ("Vehicle does not qualify for
EV charging")

Charge cost = 0.0

ELSE

IF ReqSOC <= SOC THEN

Print ("No charging Required")

Charge cost = 0.0

ELSE

Charge Needed = ReqSOC - SOC

IF SOC <= 15 AND ReqSOC >= 80 THEN

Priority = "Emergency charging priority (P1)"

ELSE IF Dis == "Y" OR (Mem == "Y" AND SOC <= 30)

THEN

Priority = "Priority charging (P2)"

ELSE

Priority = "Normal charging (P3)"

ENDIF

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IF Time $\geq$ 17.00 AND Time $\leq$ 22.00

THEN

Time Status = "Peak"

Rate = 50.0

IF Mem = "Y" AND Priority $\neq$ "Emergency Priority (PI)"

THEN

Charge Discount = 0.10

ELSE

Charge Discount = 0.0

ENDIF

ELSE

Time Status = "Off-Peak"

Rate = 35.0

IF Mem = "Y" AND Priority $\neq$ "Emergency Priority (PI)"

THEN

Charge Discount = 0.20

ELSE

Charge Discount = 0.0

ENDIF

ENDIF

Charge Cost = Charge Needed * Rate * (1.0 - Charge Discount)

ENDIF

ENDIF

ENDIF

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```
IF Hours <= 2  
    THEN  
        Base Parking = 200  
ELSE IF Hours <= 5  
    THEN  
        Base Parking = 400  
ELSE  
        Base Parking = 700  
ENDIF  
ENDIF
```

```
IF Dis == "Y"  
    THEN  
        Parking Cost = 0.0  
        Parking Discount = Base Parking  
ELSE IF Mem == "Y"  
    THEN  
        Parking Discount = Base Parking * 0.20  
        Parking Cost = Base Parking - Parking discount  
ELSE  
        Parking Discount = 0.0  
        Parking Cost = Base Parking  
ENDIF  
ENDIF
```

```
IF Hours > 8 THEN  
    Stay Message = "Long-stay warning: Please relocate  
    after charging"  
    Print (stay message)  
ELSE
```

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Stay message = "Standard parking Duration"
Print (stay message)

ENDIF

Total Payable = Charge Cost + parking cost

Total Discount = (Charge Needed * Rate * Charge Discount)
+ parking Discount

Print ("Vehicle Type:" Type)

Print ("Current Battery soc:" soc, "%")

Print ("Required Battery soc:" Req soc, "%")

Print ("Charging Units needed:" Charge Needed, "%")

Print ("Charging Priority:" Priority)

Print ("Time Status:" Time status)

Print ("Charging cost:" Charge cost)

Print ("Parking cost:" parking cost)

Print ("Discount:" Discount)

Print ("Payable amount:" Total Payable)

Print (stay message)

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Start

INPUT TYPE (E/H),
Current Battery % SOC, Required %
(Req SOC), Hours, Time,
Mem (Y/N), Dis (Y/N),
Avail (Y/N)

is station
available?
Avail = Y

No

Type = E
OR Type = H AND
SOC < 40

No

Req SOC > SOC

Yes

Charge Needed = Req SOC -
SOC

is Hybrid
Type = H

Yes

Print (charging unavail-
able, parking only)

Charge cost = 0

Print (Vehicle doesn't
require charging)

Charging cost = 0

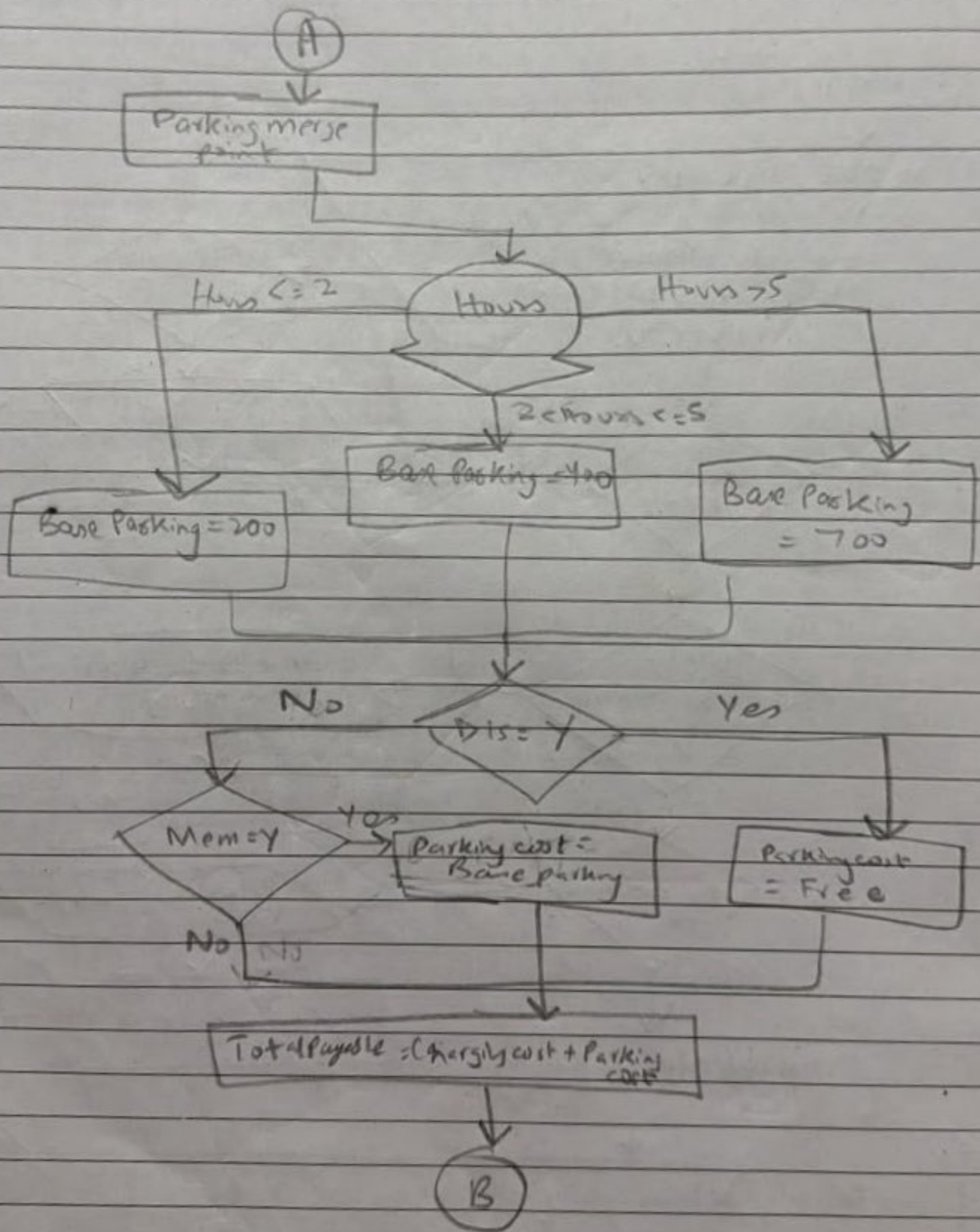
Print (No charging
required)

Charging cost = 0

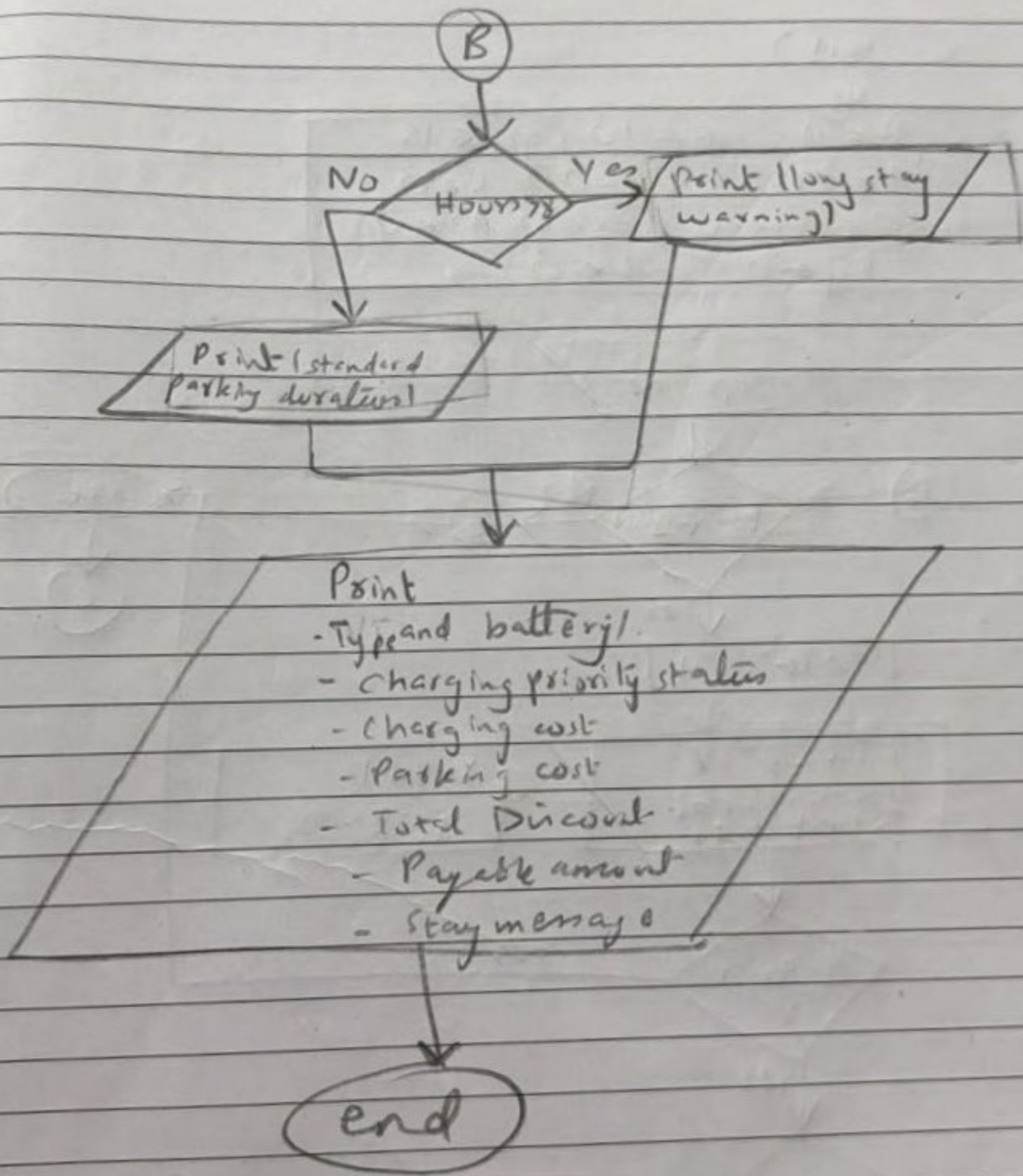
End

- Priority 1 (Emergency): If $SOC \leq 15\%$ and $Req SOC \geq 80\%$.
- Priority 2 (Priority): If $disable d = Y$ or $(Mem = Y$ and $SOC \leq 30\%)$.
- Priority 3 (Normal): Otherwise:

Dated: _____



Dated: _____



INPUTS	OUTPUT
* Vehicle Type (Type: 'E' or 'H')	* Vehicle Type (Type)
* Current Battery SOC (SOC)	* Current Battery SOC (SOC)
* Target / Required SOC (Req, SOC)	* Required Battery SOC (Req, SOC)
* Expected Parking duration (Hours)	* Units needed (charge needed)
* Current time (Time, 24-hrs format)	* Priority status message (Priority)
* Membership status (Mem: 'Y'/'N')	* Time category (Time status)
* Disabled priority status (Dis: 'Y'/'N')	* Charging Cost (Charge cost)
* Station availability (Avail: 'Y'/'N')	* Parking Cost (Parking Cost)
* Rates: Peak = 50.0, Off-Peak = 35.0	* Total Discount (Total Discount)
* Base Parking rates: 200 (≤ 2 hrs), 400 (≤ 5 hrs), 700 (> 5 hrs)	* Total payable (Total Payable)
	* Parking stay message (StayMessage)
	* Status/error message (unavailability/qualification).

PROCESSING REQUIRED	SOLUTION ALTERNATIVES
* Check station availability (Avail == "N")	1) Calculate step-by-step using nested condition blocks (current pseudocode approach)
* Validate qualification: Vehicle must	2) Modularize into separate sub-

be Electric ('E) or Hybrid ('H) with $SOC < 40$

* Calculate units needed:
 $ChargeNeeded = ReqSOC - SOC$

* Determine Priority:

P1 (Emergency), P2 (Priority),
P3 (Normal)

* Determine Time Status
and Rate: Peak (17:00 -
22:00, 50.0) or Off-
Peak (35.0)

* Calculate Charging
Discount: 10% (Peak) or
20% (Off-Peak) if member
(non-P1)

* Calculate ChargeCost =
 $ChargeNeeded * Rate * (1.0 -$
Charge Discount)

* Determine base parking fee and
+ apply disability/membership discount

* Check stay warning if Hours > 8

* Total Payable = ChargeCost +
Parking Cost

* Total Discount = (ChargeNeeded *
Rate * Charge Discount) + Parking
Discount

Functions:

ComputeChargingFee(),

ComputeParkingFee(), and

DisplayBill()

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INPUT	PROCESSING	OUTPUT
- TYPE - SOC - Req SOC - Hours - Time - Mem - Dis - Avail	Input Data: Read Type, SOC, Req SOC, Hours, Time, Mem, Dis, Avail. 2) If Avail == 'N', set Charge cost = 0.0 and display unavailability message. Else if not qualified (Type != 'E' and not (Type == 'H' and SOC < 40)) set charge cost = 0 Else, Charge Needed = Req SOC - SOC 3) Determine Priority if SOC < 15 AND Req SOC > 80 Priority = P1 (Emergency) Else if Dis == "Y" OR Mem == "Y" AND SOC < 30) Priority = P2 (Priority) Else Priority = P3 / Normal 4) Time Status, Rate and Discount: If Time >= 17 AND Time <= 22 Time status = Peak, Rate = 50 If Mem == 'Y' and Not P1, Charge Discount = 0.1 Else Time status = "Off-Peak" Rate = 35 If Mem == "Y" and Not P1 charge discount = 0.2 else 0.	- Type - SOC - Req SOC - Charge Needed - Priority - Time status - Charge Needed - Priority - Time status - Charge cost - Total discount - Total payable - Stay Manage - Status Notification

* Charge Cost = Charge Needed * Rate *

(1.0 - Charge Discount)

5) Calculate Parking Cost & Parking Discount:

* Base Parking: 200 (hours <= 2), 400 (hours <= 2), 700 (hours <= 2)

* If disc = "Y": Parking cost = 0.0, Parking discount

= Base Parking

* Else if Mem = "Y":

Parking Discount = Base Parking * 0.20, Parking

Cost = Base Parking - Parking Discount

* Else:

Parking Discount = 0.0, Parking Cost = Base Parking

6) Check Stay Duration:

* If Hours > 8, set warning stay messages
else, standard duration message

7) Calculate Totals:

* Total Payable = Charge Cost + Parking Cost

* Total Discount = (Charge Needed * Rate *
Charge Discount) + Parking Discount

8) Output Results:

Display all summary fields & bills.